1. (a)	$\dfrac{5x}{x^2+7x+12}+\dfrac{5x}{x+4}=\dfrac{5x+5x(x+3)}{(x+3)(x+4)}$	M1 A1
	$=\dfrac{5x^2+20x}{(x+3)(x+4)}=\dfrac{5x(x+4)}{(x+3)(x+4)}=\dfrac{5x}{x+3}$ *	A1*
		(3)
(b)	$y=\dfrac{5x}{(x+3)}\Rightarrow xy+3y=5x\Rightarrow 5x-xy=3y$	M1
	$\Rightarrow x=\dfrac{3y}{5-y}$ So $f^{-1}(x)=\dfrac{3x}{5-x}$	A1
	Domain $0<x<5$	A1
		(3)
(c) (i)	$f(x)=\dfrac{5x}{(x+3)}\Rightarrow (f'(x)=)\dfrac{5(x+3)-5x}{(x+3)^2}=\dfrac{15}{(x+3)^2}$	M1 A1
(ii)	States (f is an) increasing function with a suitable reason E.g. Since $(x+3)^2$ is positive	A1
		(3)
		(9 marks)

2 (a)	$\left(\frac{13}{3}, 5\right)$	B1 B1
		(2)
(b)(i)	$f(x) \ldots 5$	B1 ft
(ii)	10	B1
		(2)
(c)	Attempts to solve either $16-2x...3x-13+5 \Rightarrow$ a value or inequality in x Or $16-2x...-3x+13+5 \Rightarrow$ a value or inequality in x	M1
	Both correct critical values $2, \frac{24}{5}$	A1
	Selects inside region for their critical values	dM1
	$2 < x < \frac{24}{5}$	A1
		(4)
(d)	$a = 4, b = \frac{1}{3}$	B1 ft B1
		(2)
		(10 marks)
Alt(c)	$(11-2x)^2 ... (3x-13)^2 \Rightarrow 121-44x+4x^2 ... 9x^2-78x+169$	M1
	$5x^2-34x+48 ... 0 \Rightarrow$ a value or inequality in x	A1
	$2, \frac{24}{5}$	
	Selects inside region for their critical values	dM1
	$2 < x < \frac{24}{5}$	A1

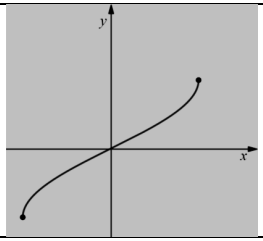
3.(a) (i)	$0 = 40 - 30e^{1-0.05t} \Rightarrow 1 - 0.05t = \ln\frac{40}{30}$	M1
(ii)	$\Rightarrow (k =)$ awrt 14.2	A1
	March 1814	A1
		(3)
(b)	Attempts $G = 40 - 30e^{1-0.05\times 70} = 37.5$ tonnes	M1 A1
		(2)
(c)	40 (tonnes)	B1
		(1)
		(6 marks)

4 (a)	$2\sin(\theta - 30°) = 5\cos\theta \Rightarrow 2\sin\theta\cos 30° - 2\cos\theta\sin 30° = 5\cos\theta$	M1
	$\div\cos\theta \qquad \Rightarrow 2\tan\theta\cos 30° - 2\sin 30° = 5$	dM1
	$\Rightarrow 2\tan\theta \times \frac{\sqrt{3}}{2} - 2\times\frac{1}{2} = 5$	A1
	$\Rightarrow \sqrt{3}\tan\theta = 6 \Rightarrow \tan\theta = 2\sqrt{3}$ *	A1*
		(4)
(b)	Attempts $\arctan 2\sqrt{3}$ ..and then subtracts $20°$	M1
	$\Rightarrow x =$ awrt $53.9°, 233.9°$	A1, A1
		(3)
		(7 marks)

5 (i)	$\int \frac{8}{(2x-3)^3}\,\mathrm{d}x = \frac{-2}{(2x-3)^2}(+c)$ $\int_2^4 \frac{8}{(2x-3)^3}\,\mathrm{d}x = \left[\frac{-2}{(2x-3)^2}\right]_2^4 = -\frac{2}{25}+2 = \frac{48}{25}$	M1 A1 dM1 A1
		(4)
(ii)	$\int x\left(x^2+3\right)^7\,\mathrm{d}x = \frac{1}{16}\left(x^2+3\right)^8 + c$	M1 A1
		(2)
		(6 marks)
Alt(i)	Let $u = 2x-3$ $\int \frac{8}{u^3}\times\frac{1}{2}\,\mathrm{d}u = -\frac{2}{u^2}(+c)$ $\int_2^4 \frac{8}{(2x-3)^3}\,\mathrm{d}x = \left[-\frac{2}{u^2}\right]_1^5 = -\frac{2}{25}+2 = \frac{48}{25}$	M1 A1 dM1 A1
Alt(ii)	Let $u = x^2+3$ $\int x\left(x^2+3\right)^7\,\mathrm{d}x = \int \frac{u^7}{2}\,\mathrm{d}u = \frac{u^8}{16}+c = \frac{1}{16}\left(x^2+3\right)^8 + c$	M1A1

6 (i)	$\left(\frac{dy}{dx} =\right)\frac{6x}{x^2-5}-8x$	M1 A1
	Stationary point when $\frac{6x}{x^2-5}-8x=0 \Rightarrow x^2=\frac{23}{4} \Rightarrow x=\frac{\sqrt{23}}{2}$ only	dM1 A1
		(4)
(ii)(a)	$\left(\frac{dy}{dx} =\right)4-24\sin x\cos x$	M1
(b)	$\left(\frac{dy}{dx} =\right)4-12\sin 2x$	dM1 A1
	Maximum gradient = 16	A1
		(4)
		(8 marks)

7.(a)	$\log_{10} M = 1.93 \log_{10} 45 + 0.684 \Rightarrow \log_{10} M = 3.8747$	M1
	$(M =)$ awrt 7500 (kg)	A1
		(2)
(b)	$\log_{10} M = 1.93 \log_{10} r + 0.684$ $\log_{10} M = \log_{10} r^{1.93} + 0.684$ $M = 10^{\log_{10} r^{1.93} + 0.684}$ $M = 10^{0.684} \times r^{1.93}$	$M = p\, r^q$ $\log_{10} M = \log_{10} pr^q$ $\log_{10} M = \log_{10} p + \log_{10} r^q$ $\log_{10} M = \log_{10} p + q \log_{10} r$ B1
	Either $p = 10^{0.684}$ or $q = 1.93$	M1
	Both $p =$ awrt 4.83 and $q = 1.93$	A1
		(3)
(c)	"p" is the mass (in kg) of a tree with radius 1 cm	B1
		(1)
		(6 marks)

8 (a)	 Shape and position	B1 **(1)**
(b)	$x = 2\sin y \Rightarrow \left(\frac{dx}{dy} =\right) 2\cos y$ and attempts to use **both** $\frac{dy}{dx} = 1 \div \frac{dx}{dy}$ and $\cos y = \sqrt{1-\sin^2 y} = \sqrt{1-...x^2}$ $\left(\frac{dy}{dx} =\right) \frac{1}{2\sqrt{1-\frac{x^2}{4}}}$ $\left(\frac{dy}{dx} =\right) \frac{1}{\sqrt{4-x^2}}$	M1 A1 A1cso
		(3)
(c)	Substitutes $x = \sqrt{2}$ into their $\frac{dy}{dx} = \frac{1}{\sqrt{4-x^2}}$ $\left(=\frac{\sqrt{2}}{2}\right)$ Finds the equation of the tangent at P $y - \frac{\pi}{4} = "\frac{\sqrt{2}}{2}"\left(x - "\sqrt{2}"\right)$ $y = \frac{\sqrt{2}}{2}x - 1 + \frac{\pi}{4}$	M1 dM1 A1
		(3)
		(7 marks)
Alt(b)	$y = \arcsin\left(\frac{x}{2}\right) \Rightarrow \left(\frac{dy}{dx} =\right) \frac{1}{2} \times \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}}$ $\left(\frac{dy}{dx} =\right) \frac{1}{\sqrt{4-x^2}}$	M1A1 A1

9 (a)	$f(x)=\left(x^3-4x\right)e^{-\frac{1}{2}x} \Rightarrow f'(x)=\left(3x^2-4\right)e^{-\frac{1}{2}x}-\frac{1}{2}\left(x^3-4x\right)e^{-\frac{1}{2}x}$	M1 A1
		(2)
(b)	$f'(0)=-4$ so equation of normal is $y=-\frac{1}{-4}x$	M1
	$y=\frac{1}{4}x$	B1
	Sets $\frac{1}{4}\cancel{x}=\cancel{x}\left(x^2-4\right)e^{-\frac{1}{2}x} \Rightarrow x^2-4=\frac{1}{4}e^{\frac{1}{2}x}$	M1
	$\Rightarrow x^2=\frac{16+e^{\frac{1}{2}x}}{4} \Rightarrow x=-\frac{1}{2}\sqrt{16+e^{\frac{1}{2}x}}$ *	A1*
		(4)
(c)	(i) $x_2=-\frac{1}{2}\sqrt{16+e^{\frac{1}{2}x-2}}=-2.0229$	M1 A1
	(ii) $\left(x=\right)-2.0226$	A1
		(3)
		(9 marks)

10.(a)	$(1+2\cos 2x)^2 = 1+4\cos 2x+4\cos^2 2x$	
	Uses $\cos 4x = 2\cos^2 2x - 1 \Rightarrow (1+2\cos 2x)^2 = 1+4\cos 2x+2\cos 4x+2$	M1
	$= 3+4\cos 2x+2\cos 4x$	A1
		(2)
(b)	$a = \dfrac{2\pi}{3}$	B1
	$\int 3+4\cos 2x+2\cos 4x \, dx = 3x+2\sin 2x+\dfrac{1}{2}\sin 4x$	M1 A1 ft
	Area $= \left[3x+2\sin 2x+\dfrac{1}{2}\sin 4x\right]_0^{\frac{2\pi}{3}} = 2\pi - \dfrac{3}{4}\sqrt{3}$	dM1 A1
		(5)
		(7 marks)